

Satyam Tiwari

AI Engineer | NLP, Machine Translation & Speech Processing | Multilingual Language Technologies

✉ itssatyam.345@gmail.com | [in LinkedIn](#) | [Portfolio](#) | ☎ +81-7040808649 | 📍 Tokyo, Japan

SUMMARY

Machine Learning / AI Engineer with 2+ years building production NLP, speech, and multilingual language systems in Python, across the full cycle from research PoC through design, development, and deployment. Hands-on with automatic speech recognition (ASR) and text-to-speech (TTS), Transformer-based multilingual embeddings, model evaluation and benchmarking, and inference optimization for real-time, low-latency processing. Reads English technical papers, builds internal evaluation tooling, and turns prototypes into customer-facing services on Linux with Docker. Business-level Japanese, effective across bilingual (JP/EN) teams.

SKILLS

NLP & Speech: Multilingual NLP, ASR (Whisper, Faster-Whisper / CTranslate2), TTS (VITS, FastSpeech2, HiFi-GAN, Conformer), Transformer Embeddings, Sequence-to-Sequence Models, Text Normalization

ML / Deep Learning: PyTorch, Transformers, scikit-learn, Model Evaluation & Benchmarking (AUROC/F1), Inference Optimization & Lightweighting, Real-Time / Low-Latency Processing

Generative AI / LLM: LLM Orchestration (Gemini, Claude, OpenAI), Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain

Backend & Data: Python, Django, FastAPI, REST APIs, PostgreSQL, SQLite, JSON/CSV, Pandas, NumPy

Cloud / DevOps: Docker, Linux, Git, Amazon Bedrock, Azure OpenAI, OpenAI & Cohere APIs

Languages: Hindi (Native), English (Fluent), Japanese (Business Level)

EXPERIENCE

Data Scientist Mar 2024 - Present
Human Resocia Co. Ltd Tokyo, Japan

AI Design Config Review System | *ICT & Social-Infrastructure System Integrator* Apr 2026 - Present

- Owned end-to-end delivery (requirements through deployment) of a generative-AI service that reviews technical design documents against device configurations, built on an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline (chunking, indexing, hybrid retrieval) over multi-format documents (PDF, PPTX, DOCX, XLSX).
- Built a **Django** backend and web application and orchestrated LLM review across **Gemini** and **Claude**, turning business requirements into a customer-facing solution; presented the system to management in Japanese.
- Diagnosed a ~20-minute document-processing bottleneck (embedding-model load on cloud-synced storage) through runtime profiling and designed a content-hash caching layer to cut repeat processing from minutes to seconds.
- Tech:** Python, Django, LangChain, ChromaDB, sentence-transformers, PyMuPDF, python-pptx, Gemini, Claude, PostgreSQL

Medical Document Similarity & EMR System | *Integrated Electronics Manufacturer (R&D Lab)* Apr 2025 - Mar 2026

- Built a **similarity-search** system over clinical records using **TF-IDF**, **cosine similarity**, and **KNN** on Transformer embeddings to retrieve and rank related documents.
- Benchmarked and optimized **multiple embedding models** (OpenAI, Cohere, Amazon Titan) through systematic offline **evaluation (AUROC/F1)** and error analysis, improving retrieval accuracy.
- Built preprocessing/evaluation pipelines and a **Streamlit** UI for EMR (JSON) management and similarity visualization (Pyvis); hybrid rule-based + ML scoring absorbed medical-domain expression variance.
- Tech:** Python, scikit-learn, Transformers, Pandas, NumPy, Amazon Bedrock, OpenAI API, Cohere API, Streamlit, Pyvis

Real-Time Speech AI (ASR/TTS) | *Integrated Electronics Manufacturer (R&D Lab)* Mar 2024 - Mar 2025

- Built a **real-time**, voice-operable system for hands-free data entry/edit/delete using **Faster-Whisper** (ASR) and **VITS** (TTS) with a Streamlit live-recording UI; reduced latency and improved transcription accuracy via optimized (CTranslate2) inference and string-matching post-processing.
- Tech:** Python, PyTorch, Faster-Whisper, VITS, ESPnet, Conformer, FastSpeech2, HiFi-GAN, Streamlit

AI Developer Intern Jan 2023 - Jun 2023
Digipodium, Inc. Lucknow, India

- Built a **voice-based email system** for visually impaired users (Google Speech & Gmail APIs, Django, JavaScript); ~90% ASR accuracy, ~50% task-time reduction.

AI Developer Intern Jun 2022 - Sept 2022
Digipodium, Inc. Lucknow, India

- Built a **real-time object recognition system** using TensorFlow and OpenCV with audio feedback; labeled 1,000+ training images.

EDUCATION

Amity University, Lucknow India
Master of Computer Applications (Information Technology) | GPA: 8.45 2021 - 2023